Plane Geometry

ANSWER KEY

Section 1: Angles & Parallel Lines

1 Two parallel lines are intersected by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 45)^\circ$, what is the value of x?

a. 24 ✓

b. 30

c. 36

d. 40

Consecutive interior angles are supplementary:
$(3x + 15) + (2x + 45) = 180 \Rightarrow 5x + 60 = 180 \Rightarrow 5x = 120 \Rightarrow x = 24$.

2 In a triangle, the ratio of the three interior angles is $2 : 3 : 4$. What is the degree measure of the largest angle?

a. 60°

b. 75°

c. 80° ✓

d. 90°

Sum of angles $= 180^\circ$. Total parts $= 2 + 3 + 4 = 9$. Each part $= \frac{180}{9} = 20^\circ$. The largest angle is $4 \times 20^\circ = 80^\circ$.

3 What is the measure of each interior angle of a regular octagon?

a. 120°

b. 135° ✓

c. 140°

d. 144°

Formula for interior angle of regular n-gon: $\frac{(n-2) \times 180^\circ}{n}$. For $n = 8$: $\frac{6 \times 180^\circ}{8} = \frac{1080^\circ}{8} = 135^\circ$.

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Section 2: Triangles, Similarity & Pythagorean Theorem

4 In a right triangle, the hypotenuse has length 25 cm and one leg has length 7 cm. What is the length of the other leg?

a. 20 cm

b. 24 **cm** ✓

c. 22 cm

d. 18 cm

By the Pythagorean theorem: $a = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ cm.

5 Two similar triangles have corresponding side lengths in the ratio $3:5$. If the area of the smaller triangle is 36 square units, what is the area of the larger triangle?

a. 60

b. 80

c. 100 ✓

d. 125

Ratio of areas equals the square of the ratio of sides: $\frac{\text{Area}_1}{\text{Area}_2} = (\frac{3}{5})^2 = \frac{9}{25}$. Then $\frac{36}{\text{Area}_2} = \frac{9}{25} \Rightarrow \text{Area}_2 = 36 \times \frac{25}{9} = 100$.

6 In a 30°-60°-90° right triangle, the length of the hypotenuse is $16\sqrt{3}$. What is the length of the side opposite the 60° angle?

a. 12

b. 24 ✓

c. $8\sqrt{3}$

d. 16

Side opposite 30° is half the hypotenuse: $\frac{16\sqrt{3}}{2} = 8\sqrt{3}$. Side opposite 60° is $(8\sqrt{3})\sqrt{3} = 8 \times 3 = 24$.

Section 3: Quadrilaterals & Polygons

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7 A trapezoid has parallel bases of lengths 14 cm and 26 cm and an altitude of 9 cm. What is the area of the trapezoid?

a. 160 cm^2

b. 180 **cm^2** ✓

c. 200 cm^2

d. 220 cm^2

Area $= \frac{b_1 + b_2}{2} \times h = \frac{14 + 26}{2} \times 9 = \frac{40}{2} \times 9 = 20 \times 9 = 180$ cm^2.

8 The diagonals of a rhombus have lengths 16 inches and 30 inches. What is the perimeter of the rhombus?

a. 60 inches

b. 64 inches

c. 68 **inches** ✓

d. 72 inches

Diagonals bisect each other perpendicularly, creating four right triangles with legs $\frac{16}{2} = 8$ and $\frac{30}{2} = 15$. Side length $s = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. Perimeter $= 4 \times 17 = 68$ inches.

9 A rectangle has a perimeter of 48 meters and a length that is twice its width. What is the area of the rectangle?

a. 96 m^2

b. 128 **m^2** ✓

c. 144 m^2

d. 160 m^2

$P = 2(L + W) = 2(2W + W) = 6W = 48 \Rightarrow W = 8$. Then $L = 16$. Area $= 16 \times 8 = 128$ m^2.

Section 4: Circles, Arcs & Sectors

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10 A circle has a radius of 12 cm. What is the area of a sector formed by a central angle of 60°?

a. 12π cm^2

b. 24π **cm^2** ✓

c. 36π cm^2

d. 48π cm^2

Sector area $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60}{360} \times \pi(12^2) = \frac{1}{6} \times 144\pi = 24\pi$ cm^2.

11 In a circle of radius 10 inches, what is the length of an arc intercepted by a central angle of $\frac{3\pi}{5}$ radians?

a. 4π inches

b. 6π **inches** ✓

c. 8π inches

d. 10π inches

Arc length formula in radians: $s = r\theta = 10 \times \frac{3\pi}{5} = 6\pi$ inches.

12 An inscribed angle intercepts an arc of 110° on a circle. What is the degree measure of the inscribed angle?

a. 55° ✓

b. 70°

c. 110°

d. 220°

An inscribed angle is equal to half the measure of its intercepted arc: $\frac{110^\circ}{2} = 55^\circ$.

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